Explain how scientific knowledge is validated and refined, including the role of publication and peer review

Learning intention

- explain how scientific knowledge is validated and refined, including the role of publication and peer review.

Teach and model

- exploring how astronomer Vera Rubin’s discovery of the existence of dark matter was validated.
- examining why there are different climate change models used by scientists when there is a climate change consensus among scientists.
- exploring the role of large data sets and statistical analysis in validating scientific findings, such as Gregor Mendel’s experiments with pea plants.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.